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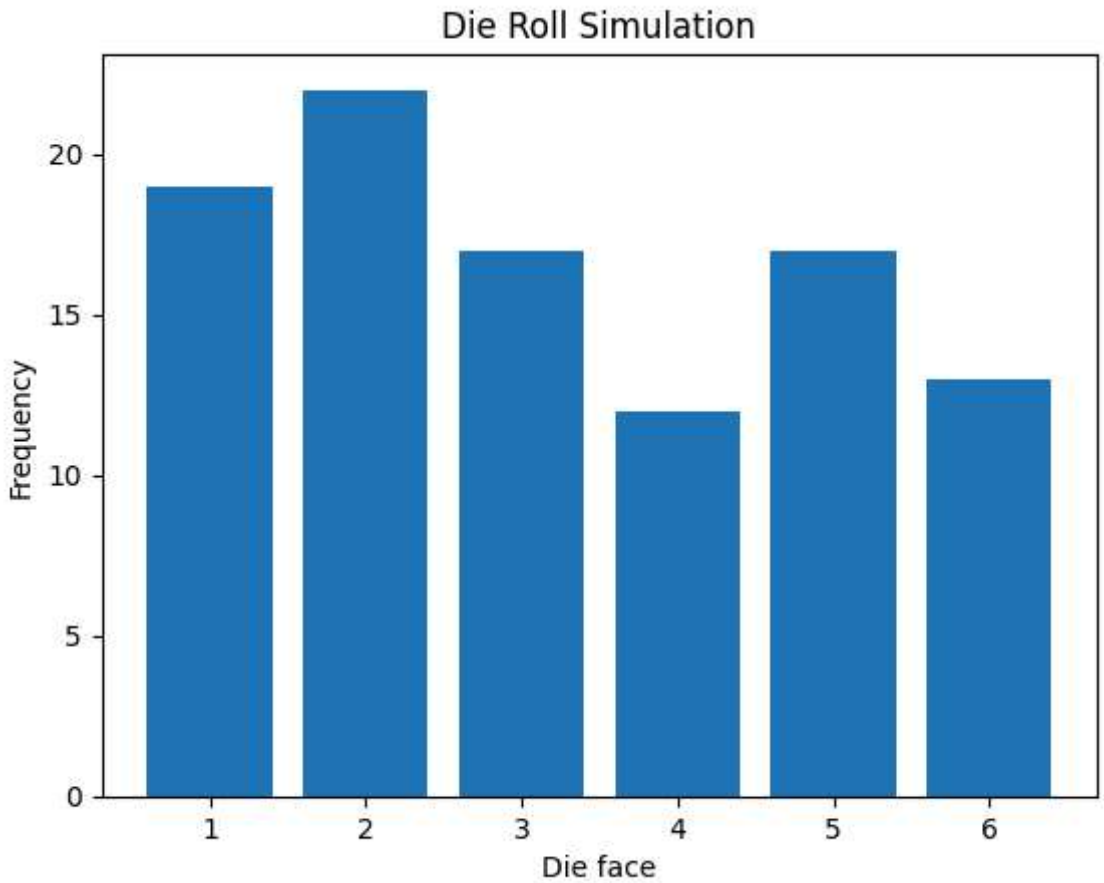
In [ ]:
```
#1. Rolling a dice 100 times
import random
import matplotlib.pyplot as plt
```

In [17]:
```
num_rolls = 100
outcomes = [] #blank list to store outcomes
#simulation
for _ in range(num_rolls):
    outcomes.append(random.randint(1,6))

#Calulating probability
faces = [1,2,3,4,5,6]
frequency = []
probability = []
print("Face\t Frequency\t Probability")

for face in faces:
    freq = outcomes.count(face)
    prob = freq/num_rolls
    probability.append(prob)
    frequency.append(freq)
    print(face, "\t", freq, "\t\t", prob)
```

```
Face     Frequency       Probability
1        19              0.19
2        22              0.22
3        17              0.17
4        12              0.12
5        17              0.17
6        13              0.13
```

In [ ]:
```
#plotting Bar graph
plt.bar(faces, frequency)
plt.xlabel("Die face")
plt.ylabel("Frequency")
plt.title("Die Roll Simulation")
plt.xticks(faces)
plt.show()
```

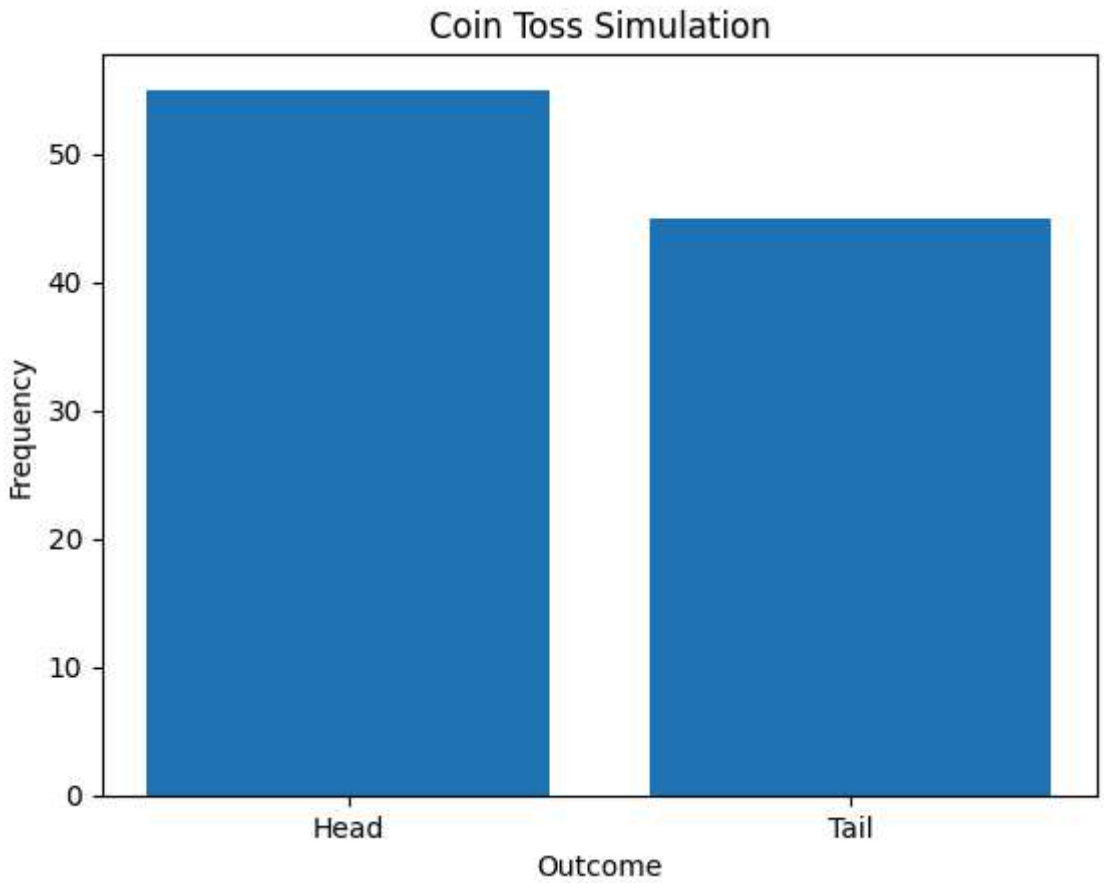


In [ ]:
```
#2. Tossing a coin 100 times
import random
import matplotlib.pyplot as plt

num_tosses = 100
outcomes = []

#Simulation of tossing of coin
for _ in range(num_tosses):
    outcomes.append(random.choice(['H','T']))

#Calculating Probability
head_count = outcomes.count('H')
tail_count = outcomes.count('T')
prob_head = head_count/ num_tosses
prob_tail = tail_count/ num_tosses
print("Outcome\t Frequency\t Probability")
print(f"Head\t {head_count}\t\t {prob_head: 0.2f}")
print(f"Tail\t {tail_count}\t\t {prob_tail: 0.2f}")
```

```
Outcome  Frequency       Probability
Head     55              0.55
Tail     45              0.45
```

In [33]:
```
#Plotting Bar graph
labels = ["Head", "Tail"]
frequency = [head_count, tail_count]
plt.bar(labels, frequency)
```

```
plt.xlabel("Outcome")
plt.ylabel("Frequency")
plt.title("Coin Toss Simulation")
plt.show()
```



In [ ]: